

Yiwen GENG

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EDUCATION

RICE UNIVERSITY

HOUSTON, TX

Master of Science, Electrical & Computer Engineering — GPA 3.91/4.0

Aug 2021 – Aug 2025

Coursework: Advanced VLSI Design (A), Computer Systems Architecture (A+), Statistical Signal Processing, Embedded ML (A)

TSINGHUA UNIVERSITY

BEIJING, CHINA

Bachelor & Master of Engineering, Integrated Circuit Engineering

Aug 2015 – Jun 2021

Coursework: Process of VLSI Fabrication & Equipment (A-), Semiconductor Devices, Digital & Analog IC Design

TECHNICAL SKILLS

Programming Languages & Tools: Verilog, Python, C++, Git, Bash

Software: Virtuoso, ModelSim, Matlab, Cadence, Synopsys (Design Compiler, IC Compiler)

Hardware: Agilent B1500A parameter analyzer (DC sweep), Cascade Summit 11000 probe station, Nextest Maverick II memory tester, Mixed Domain Oscilloscope, Soldering, BER Tester, Arista Switch, Programming board

EXPERIENCE

INFRAEO INC.

ROUND ROCK, TX

Embedded Systems Engineer

Jan 2026 – Present

- Build and execute the qualification test suite for each module build — define test scope and pass/fail criteria, and verify measured device behavior against vendor characterization data and IEEE 802.3 / OIF-CEI electrical specifications.
- Own the debug loop on BER and factory test data — identify failure modes and isolate root cause across silicon, firmware, and physical layer using BERT and oscilloscope measurements (eye diagrams, jitter, signal integrity); define corrective actions and trace them to verify effectiveness in outgoing-quality validation; reproduce and document anomalies for the silicon vendor and drive closure with vendor, hardware, and firmware teams.
- Bring up and debug firmware and hardware across switch, BERT, and programming boards; build Python command-line tools that drive BERT and programming boards over USB and I2C, automating data acquisition, test execution, and configuration, and extending them into a hardware-in-the-loop regression framework.
- Proposed and built **SPELL** (Spec-to-Executable via LLM), an LLM-driven auto-correction system for 200G to 1.6T copper and optical modules, implemented as a custom compiler that grounds every check in the authoritative CMIS specification and automatically corrects register-level anomalies such as EEPROM malprogramming and configuration mismatches.

XPERF INC.

ROUND ROCK, TX

Software Engineer

Sep – Dec 2025

- Deployed a GPU observability platform (NVIDIA DCGM, Prometheus, Grafana) on GKE, monitoring utilization/memory/thermals across multi-tenant ML workloads with automated failure detection.
- Migrated to a push-based telemetry architecture, cutting GPU failure-detection latency minutes→seconds; automated GPU VM provisioning on GCP (deployment time -40%); developed and versioned all code in a Git-based workflow.

RESEARCH & PROJECTS

On-chip Layer-wise Regression to Compensate Array Non-idealities — RRAM Neuromorphic Chips (Tsinghua LEMON Lab; *first-author, DATE 2021*)

- **Device & array characterization:** DC-sweep on an Agilent B1500A; continuous resistance-change characterization of a **1K-bit array** (Cascade 1100 probe station, lead probe cards, Nextest Maverick2 tester); built a Python analog-RRAM model capturing measured non-idealities.
- **CIM method & hardware:** layer-wise regression compensating array non-idealities (+10% ResNet-18 accuracy at 25% of the operations); implemented the data-input path on Xilinx Zynq + AXI in Verilog, and rewrote PE_update / Send_upper_data in C++ and the PE_switch_path block in Verilog.

Speech Signal Processing & Statistical Data Analysis (Rice — Digital Health Lab, Prof. Ashutosh Sabharwal)

- Drove data-science research on traumatic load and physiological markers in pregnancy; hosted data collection, audio annotation, and Python cleaning / feature extraction.
- Reduced speaker-diarization error by **5–16% vs a Pyannote baseline** at **75% scenario-classification accuracy** (transformer-based multi-task model); ran **multiple correspondence analysis** relating social variables (education, occupation), physiological markers (sleep, HRV), and trauma scores (ACEs).

Si:P Doping Prediction for CMOS Devices (first-principles / DFT) — forming/ionization energy and channel strain across phosphorus-doping patterns for device-design trade-offs.

Digital design: 16-bit HSPICE multiplier (Booth + Wallace-tree); 32-bit pipelined MIPS on FPGA.

ADDITIONAL INFORMATION

Publications & patents: 6 papers (first-author DATE 2021; IEEE MTT-S NEMO 2026 accepted; manufacturing-defect detection, Fundamental Research 2022); 3 Chinese patents and 1 U.S. patent. **Languages:** English, Mandarin.